

The Uncertain Exorbitant Privilege and Duty *

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Abstract

This paper examines how shocks to exchange rate and macroeconomic uncertainty affect the United States' net foreign asset (NFA) position. I employ a structural VAR that combines external instruments, narrative (event) inequalities, and shock-dependent restrictions to address endogeneity in uncertainty measures and their links to net portfolio flows. I find that both exchange rate and macroeconomic uncertainty shocks reduce the United States' NFA deficit, with macroeconomic uncertainty exerting a larger and more persistent effect than exchange rate uncertainty. The NFA improvement is accompanied by greater macroeconomic fluctuations in the short run but dampened volatility at longer horizons. In addition, macroeconomic uncertainty raises exchange rate volatility, whereas higher exchange rate uncertainty tends to modestly dampen macroeconomic uncertainty. These results are consistent with the convenience-yield and dominant-currency pricing literatures, heightened volatility rebalances foreign demand toward United States safe assets and transmits through global terms of trade and financing conditions.

Keywords: Uncertainty, Net Foreign Asset Position, Exchange Rate, Capital Flows

JEL Codes: E44, F31, F44, C32, C38

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1 Introduction

Feedback effects in capital flow behavior endogenously determine relationships between domestic and foreign asset prices across internationally integrated economies. Recent exchange rate liberalization has deepened financial integration—facilitating global investment, enhancing liquidity, and enabling freer capital mobility. However, this integration comes with costs: capital inflows and outflows introduce volatility in asset prices, which is reflected in fluctuations in exchange rates.

In recent years, the United States (US) has benefited from sustained global demand for its liquid financial instruments—particularly safe assets such as Treasury bills and government bonds. This demand has enabled the US to maintain trade and current account deficits while keeping interest rates relatively low, (Gourinchas and Rey, 2007, 2014, 2022). These inflows are offset by substantial outbound portfolio investment by US investors, who earn higher returns reflecting foreign risk premia. This dynamic underpins the so-called “exorbitant privilege,” where the US achieves high returns on foreign assets at relatively low cost, coupled with the “exorbitant duty” of providing global safe assets—especially during periods of heightened financial stress.

However, under conditions of elevated uncertainty, domestic risk premia rise alongside exchange rate volatility. In such cases, foreign investors may rebalance their portfolios, triggering a sudden stop or retrenchment of capital flows. These dynamics raise questions about the persistence of the United States’ exorbitant privilege and duty, and their implications for global financial stability.

In this paper, I examine how exchange rate and macroeconomic uncertainty affect gross portfolio capital flows to the US, with a particular focus on their implications for the Net Foreign Asset (NFA) position in long-term assets. I build on the frameworks of Gabaix and Maggiori (2015), Farhi and Gabaix (2016), Koijen and Yogo (2020), and Camanho, Hau, and Rey (2022), which study the links between capital flows, asset demand, and exchange rate movements. Specifically, I aim to disentangle the effects of global foreign exchange volatility during perceived “rare disaster” episodes, which often trigger rebalancing by both domestic and foreign investors. This helps determine whether capital inflows to the US remain persistent during episodes of elevated macroeconomic or exchange rate uncertainty.

Consistent with the literature on capital flows, I expect that periods of heightened uncertainty induce inflows to the US in search of safe assets—such as Treasuries, agency bonds, and investment-grade corporate debt—thereby amplifying the country’s exorbitant duty. At the same time, precautionary behavior leads to a sudden stop in outward equity flows, as domestic investors rebalance their portfolios and forgo their privilege. I find evidence that

rising uncertainty affects the NFA position through a contraction in foreign asset holdings by domestic investors that exceeds the reduction in US asset holdings by foreign investors. Macroeconomic uncertainty shocks lead to reduced foreign demand for both US bonds and equities. By contrast, domestic investors maintain demand for foreign bonds but reduce their exposure to foreign equities. The effects of exchange rate uncertainty are asymmetric: foreign investors do not reduce demand for US safe assets, while US investors rebalance away from foreign markets.

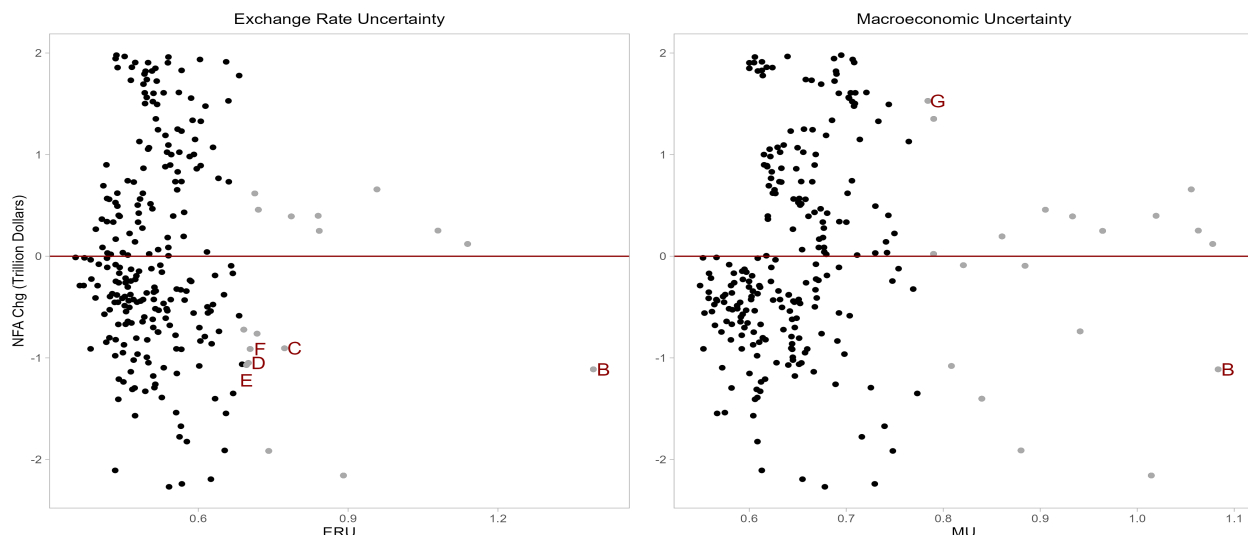
The empirical framework follows the Structural Vector Autoregressive (SVAR) models of Ludvigson, Ma, and Ng (2017) and Ludvigson, Ma, and Ng (2021), where uncertainty is treated as an endogenous variable rather than an exogenous shock. I use the Exchange Rate Uncertainty (ERU) index developed by Fernandez-Mejia (2022) to capture the role of currency volatility in portfolio decisions. Macroeconomic uncertainty is proxied using the index from Jurado, Ludvigson, and Ng (2015), based on US macroeconomic data. As argued by Ramey (2016), macroeconomic uncertainty likely exhibits a contemporaneous relationship with other macro variables, particularly those with high persistence. Following Bloom (2009a), Bloom (2014), and Alfaro, Bloom, and Lin (2023), investment dynamics are expected to exhibit strong feedback with uncertainty shocks, as firms tend to delay or scale back expenditures during such periods, amplifying future volatility.

The identification strategy recovers structural shocks without imposing traditional short- or long-run exclusion restrictions. Many SVAR identification schemes rely on assumptions that may not hold empirically (Kilian, Plante, and Richter, 2022). External instruments (e.g., Stock and Watson, 2012; Montiel Olea, Stock, and Watson, 2021) face challenges due to the high correlation between uncertainty and other macroeconomic shocks (Berthold, 2023). Similarly, sign and inequality restrictions (e.g., Uhlig, 2004) can suffer from identification fragility, as shown in Wolf (2022). To address these limitations, I apply a combination of narrative identification and shock-dependent restrictions, as in Antolín-Díaz and Rubio-Ramírez (2018) and Ludvigson et al. (2017), using critical high-uncertainty episodes to discipline the identification. This approach avoids the pitfalls of overreliance on short- or long-run restrictions and improves differentiation between macroeconomic and exchange rate uncertainty shocks. To the best of my knowledge, this is the first paper to quantify the impact of capital flows within a model that endogenizes both uncertainty and asset flows.

One of this paper’s key contributions is the use of exchange rate uncertainty to study fluctuations in capital flows to and from the US. Fernandez-Mejia (2022) shows that although ERU periods often overlap with macroeconomic uncertainty, the two indices capture distinct dynamics. As shown in Figure 1, certain events—such as the September 2001 attacks—are visible in macroeconomic uncertainty indices (Jurado et al., 2015; Baker, Bloom, and Davis,

2016) but not in ERU.¹

Figure 1. Uncertainty Indices and US Net Foreign Asset Position



Note: The plot shows changes in the monthly NFA position for the US from January 1999 to December 2019, alongside measures of exchange rate and macroeconomic uncertainty. Grey dots mark observations exceeding the 90th percentile of uncertainty. Red-labeled points correspond to key historical events: B (October 2008, post-Lehman default), C (September 2011, start of European debt crisis and Operation Twist), D (January 2015, end of Swiss franc peg and ECB QE), E (August 2015, Chinese yuan devaluation), F (January 2016, negative rates in Japan and Chinese growth concerns), G (September 2001, US market crisis). *Sources:* Author’s calculations; Fernandez-Mejia (2022); Jurado et al. (2015); Treasury International Capital (TIC) System.

Economies experiencing elevated uncertainty face heightened volatility in capital flows and are prone to extreme episodes, as frequent portfolio rebalancing destabilizes exchange rates. As the dominant currency issuer and anchor of the global financial system, the US occupies a unique position (Gopinath, Boz, Casas, Díez, Gourinchas, and Plagborg-Møller, 2020; Gopinath and Itskhoki, 2022). Persistent demand for US safe assets during both booms and busts generates a convenience yield (Caballero, Farhi, and Gourinchas, 2017; Jiang, Krishnamurthy, and Lustig, 2021), supporting its role as a global financial intermediary. This has afforded the US the exorbitant privilege of running external deficits while maintaining stable demand for its liabilities—effectively acting as the world’s venture capitalist by issuing safe assets and reinvesting globally. However, under high exchange rate uncertainty, demand for US assets may weaken. Domestic investors may increasingly perceive foreign markets as riskier and reallocate toward higher-return domestic assets, triggering a retrenchment of outflows and further compressing the NFA deficit (Atkeson, Heathcote, and Perri, 2022).

¹In Appendix A3 of Fernandez-Mejia (2022), the author shows that exchange rate uncertainty is more closely correlated with FX volatility (e.g., Menkhoff, Sarno, Schmeling, and Schrimpf, 2012) than with the VIX, and it is less correlated with measures of financial, real, or macroeconomic uncertainty.

Volatile capital flows have broader implications for financial stability. Central banks and private firms, especially those with foreign exchange-denominated liabilities, face heightened balance sheet risks when capital structure mismatches are exposed (Bruno and Shin, 2015b; Salomao and Varela, 2022). This fragility constrains liquidity, raises risk premia, and feeds back into uncertainty. Exchange rate and macroeconomic uncertainty thus destabilize capital flows, impair investment decisions, and amplify the demand for safe assets as a form of insurance (Caballero, 2016; Akinci, Kalemli-Özcan, and Queralto, 2022; Bloom, 2009b).

Literature Review The role of uncertainty in shaping macroeconomic dynamics has long attracted interest in the literature, recognizing how expectations and shocks influence agents’ decisions. Since the global financial crisis of 2007–2008, research has increasingly focused on measuring uncertainty and its implications for economic activity. The paper by Bloom (2009b) was one of the first to analyze uncertainty in relation to stock market volatility. They used the Chicago Board Options Exchange’s (CBOE) Volatility Index (VIX) and the difference between forecasted and realized firm earnings. Other measures of uncertainty rely on news-based indices, such as the Economic Policy Uncertainty (EPU) index of Baker et al. (2016), or on forecast error distributions, as in Rossi and Sekhposyan (2015). Jurado et al. (2015) proposed a new measure of uncertainty derived from a VAR with factors that forecast different macroeconomic variables, from which they subtract realized values and calculate uncertainty as conditional volatility.

Recent research has implemented different methodologies to assess uncertainty and investigate how it influences capital flows. The VAR methodology is widely used for uncertainty analysis because it incorporates uncertainty as part of a system of endogenous variables to understand its shocks. Rey (2015) and Bruno and Shin (2015a) demonstrated that an increase in the VIX has negative consequences for capital flows, leading to decreased inflows into the country. Bacchiocchi, Bastianin, Missale, and Rossi (2020) used an SVAR with mixed-sampling frequency and found that the outcomes may differ based on the frequency used, as higher periodicity shows larger reductions of flows to the US than lower ones. Mandalinci and Mumtaz (2019) analyzes the impact of both regional and global variations on capital flows using a Factor-Augmented VAR (FAVAR), incorporating the VIX and regional uncertainty measures from Mumtaz and Theodoridis (2017). This study contributes to the existing literature by introducing a novel framework to analyze the effects of uncertainty shocks on capital flows, emphasizing both the net position and gross capital flows. In the model, uncertainties arise endogenously rather than being conditioned by variable ordering.

I use the SVAR model of Ludvigson et al. (2021), which takes different types of uncertainty—financial and macroeconomic—and estimates their shocks as endogenous. Early versions of the paper initiated the debate on whether uncertainty should be treated as an

endogenous or exogenous shock and on the use of instrumental variables. Caldara, Fuentes-Albero, Gilchrist, and Zakrajšek (2016) uses an SVAR with a penalty function to distinguish between financial and macroeconomic shocks, finding that the worst types of shocks combine both. Bonciani and Ricci (2020) calculates financial uncertainty using 1,000 realized volatilities of different countries and takes the first principal component. They use linear projections to estimate the effects of different variables. Caggiano, Castelnuovo, and Figueres (2020) estimates the impact of uncertainty using October 1987 and September 2008 as shock events. Carriero, Mumtaz, Theodoridis, and Theophilopoulou (2015), Carriero, Clark, and Marcellino (2018), and Carriero, Clark, and Marcellino (2020) propose using external instruments within the Proxy-VAR framework of Stock and Watson (2012) and Mertens and Ravn (2013), which has the advantage of avoiding measurement error bias. Angelini, Bacchiocchi, Caggiano, and Fanelli (2019) and Angelini and Fanelli (2019) use variations of that methodology and present applications showing that uncertainty behaves as an exogenous shock to domestic output. Carriero, Clark, and Marcellino (2021) reconcile these strands of the literature by using a VAR with stochastic volatility that accounts for the effects of uncertainty on both mean and variance.

The assessment of the impact of exchange rate uncertainty on capital flows involves a thorough examination of its influence on significant movements in gross capital flows, including surges, stops, flights, and retrenchments. The analysis of these extreme movements was initially carried out by Forbes and Warnock (2012) and further expanded upon in Forbes and Warnock (2021), which builds upon Calvo (1998) to characterize their timing. Other authors, such as Schmidt and Zwick (2015), have also explored these extreme movements in relation to uncertainty, using a panel with various uncertainty measures to analyze different instances of extreme capital flows in the euro area. Wang and Yan (2021) use a dynamic panel with quantile regression to evaluate the impact of push and pull factors on capital flows, finding that the VIX affects both quantiles by reducing the level of flows. In this paper, I examine the changes in each component of gross capital flows and the influence of uncertainty on the portfolio decisions of international investors, filling the gap regarding the effects produced by macroeconomic and exchange rate uncertainty.

2 Econometric Model

I follow the identification approach of Ludvigson et al. (2017) and Ludvigson et al. (2021), which uses narrative (event) restrictions—augmented where relevant with sign restrictions and external instruments—to recover uncertainty shocks. The design delivers set identification rather than point identification, acknowledging model uncertainty and yielding a set of

impulse responses consistent with the data and imposed restrictions.²

Let y_t be an $N \times 1$ vector. The reduced-form VAR(p) is

$$y_t = \sum_{i=1}^p A_i y_{t-i} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $\{A_i\}_{i=1}^p$ are autoregressive coefficient matrices and Σ is the variance–covariance matrix of reduced-form innovations u_t . The structural representation is

$$u_t = B\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I_N), \quad (2)$$

so that $\Sigma = BB'$. Without additional restrictions, identification is incomplete: the mapping from Σ to B is many-to-one. We can write the covariance-matching condition as

$$\bar{g}_\Sigma(B) \equiv \text{vech}(\Sigma) - \text{vech}(BB') = 0. \quad (3)$$

This delivers only $N(N+1)/2$ equations for the N^2 elements of B , leaving $N(N-1)/2$ free parameters. Equivalently, for any Cholesky factor P with $\Sigma = PP'$, the admissible impact matrices are $B = PQ$ for orthonormal Q ($QQ' = I_N$). I draw a large number of random orthonormal matrices Q using QR decompositions and set $B = PQ$ to span the identified set.³ I will follow Antolín-Díaz and Rubio-Ramírez (2018) and Ludvigson et al. (2021) and impose narrative restrictions on shocks on significant events, precisely where there is high Uncertainty. I will define the narrative restrictions on the uncertainty index data and historical events that will allow us to determine its effects robustly.

Event Inequality Constraints

Event (narrative) inequalities discipline Q by exploiting historically high-uncertainty episodes in macroeconomic and exchange rate uncertainty⁴. This strategy aligns identification with salient data features and avoids ad hoc short- or long-run zero restrictions. Formally, let $\varepsilon_{M,t}$ and $\varepsilon_{ER,t}$ denote the structural shocks to macroeconomic and exchange rate uncertainty,

²See Kilian and Lütkepohl (2017) for an overview of set- versus point-identified SVARs.

³Following Ludvigson et al. (2021), I generate 1.5 million draws of Q to approximate the identified set of impulse responses.

⁴Using the identified episodes from the Financial Uncertainty calculated by Ludvigson et al. (2021) may be attractive. However, as shown in Fernandez-Mejia (2022), and in Appendix A, the estimated events for the ERU and the Financial Uncertainty are different. The ERU captures shocks from external events that might not capture the factor of their financial assets

respectively, and $\varepsilon_{NFA,t}$ the shock to the NFA block. I impose:

- Event 1 ($\bar{g}_1$) : $\varepsilon_{M,t} \geq \delta_1$ at $t = \text{Sep 2001}$,
- Event 2 ($\bar{g}_2$) : $\varepsilon_{ER,t} \geq \delta_2$ or $\varepsilon_{M,t} \geq \delta_3$ at $t = \text{Oct 2008}$,
- Event 3 ($\bar{g}_3$) : $\varepsilon_{ER,t} \geq \delta_4$ at $t = \text{Sep 2011}$,
- Event 4 ($\bar{g}_4$) : $\varepsilon_{ER,t} \geq 0$ at $t = \text{Jan 2015}$,
- Event 5 ($\bar{g}_5$) : $\varepsilon_{ER,t} \geq \delta_5$ at $t = \text{Oct 2015}$,
- Event 6 ($\bar{g}_6$) : $\varepsilon_{ER,t} \geq \delta_6$ at $t = \text{Jan 2016}$,
- Event 7 ($\bar{g}_7$) : $\sum_{t \in \mathcal{T}} \varepsilon_{NFA,t} \leq 0$ for $\mathcal{T} = \{\text{Sep 2008}, \dots, \text{Jun 2009}\}$.

Thresholds δ_k correspond to the 90th or 95th percentile of the respective shock distributions, as detailed below.

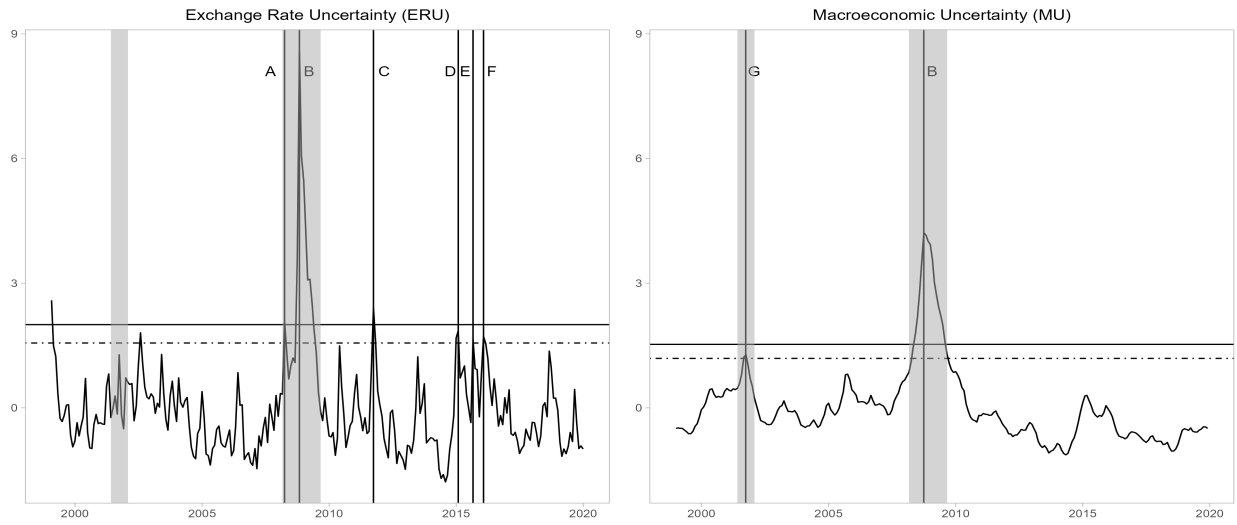
Figure 2 reports the identified episodes using thresholds at 1.65 and 1.28 standard deviations (approximately 95th and 90th percentiles) above the mean of macroeconomic and exchange rate uncertainty. Exchange rate uncertainty spikes are typically smaller in magnitude than macroeconomic uncertainty spikes, implying weaker a priori effects on the NFA. For macroeconomic uncertainty, Events $\bar{g}_1$ and $\bar{g}_2$ coincide with the September 2001 terrorist attacks and the Lehman Brothers bankruptcy, respectively—episodes used by Ludvigson et al. (2017, 2021) and Berthold (2023).

Additional events target exchange rate uncertainty and the post-GFC NFA dynamics. Event $\bar{g}_3$ captures the September 2011 escalation of concerns about European sovereign credit risk. During this period, the Federal Reserve implemented Operation Twist—selling short-term and purchasing long-term Treasuries—to put downward pressure on long-term yields, inducing portfolio rebalancing toward Treasuries. Elevated safe-asset demand and heightened uncertainty jointly sustain exchange rate uncertainty in this window.

Event $\bar{g}_4$ corresponds to the European Central Bank’s expansion of its Asset Purchase Programme (APP) and coincides with the Swiss National Bank’s removal of the euro floor. The former operates through an interest-rate and portfolio-rebalancing channel that alters asset valuations and foreign demand (see Benigno, Canofari, Di Bartolomeo, and Messori, 2023). The latter generated exchange rate volatility by shifting the liquidity properties of a safe-haven currency (the Swiss franc) and by ending foreign-exchange reserve accumulation associated with maintaining the floor (Amador, Bianchi, Bocola, and Perri, 2020; Breedon, Chen, Ranaldo, and Vause, 2023).

Event $\bar{g}_5$ —China’s devaluation of the renminbi (RMB)—triggered pronounced financial market fluctuations. Repricing of a highly traded currency issued by a major holder of

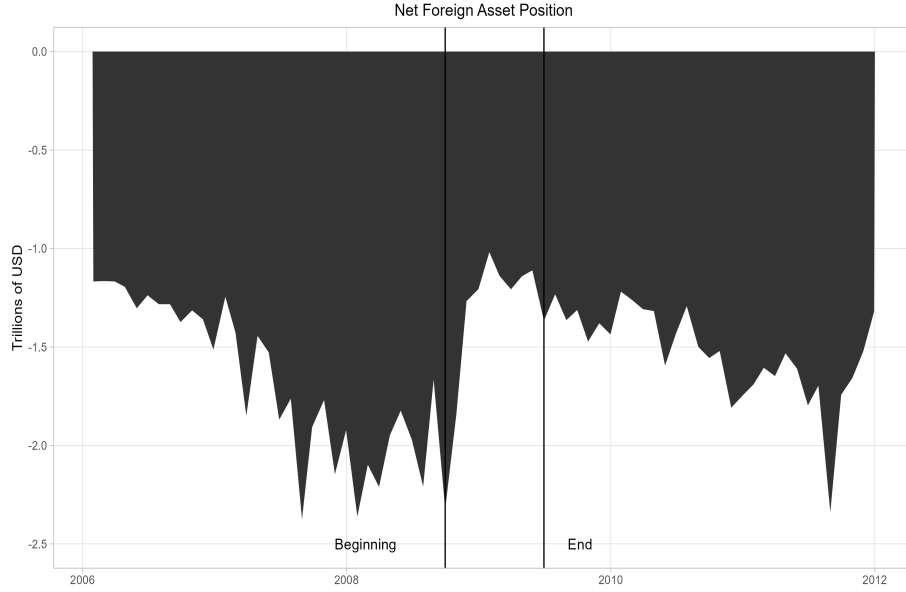
Figure 2. Exchange Rate and Macroeconomic Uncertainty over Time



Note: The figure reports ERU and macroeconomic uncertainty standardized to zero mean. The solid horizontal line is 1.65 standard deviations above the mean; the dashed line is 1.28 standard deviations. Vertical markers indicate peak months used as event constraints. A: Federal Reserve announces the Term Auction Facility; B: Lehman default; C: onset of the European sovereign debt crisis and Operation Twist (Sep 2011); D: Swiss National Bank ends the euro peg and the ECB expands asset purchases (Dec 2014/Jan 2015); E: China devalues the renminbi (Aug 2015); F: negative policy rates in Japan and concerns about China's growth (Jan 2016); G: September 2001 U.S. market disruption. *Sources:* Ludvigson et al. (2021), Fernandez-Mejia (2021), and author's calculations.

United States assets affects capital flows directly and, via terms-of-trade changes for a central economy in global trade, propagates to current accounts and exchange rates worldwide. Event $\bar{g}_6$ marks the introduction of negative policy rates in Japan and heightened concerns about China’s growth outlook. As both economies are large holders of United States assets and the yen and RMB have sizable FX turnover shares, these developments influence capital flows and exchange rate dynamics.

Figure 3. NFA and the Event $\bar{g}_6$ Restriction, 2006–2012



Note: The figure reports the NFA position in long-term assets from 2006 to 2012. The crisis window used for the cumulative restriction runs from September 2008 to June 2009 (NBER recession trough in June 2009).
Sources: Author’s calculations and TIC data.

The final restriction imposes that the cumulative effect of the crisis shock and tighter United States financial conditions is negative for the NFA. In the capital flows literature, United States shocks function as pull factors—tightening global financial conditions, weakening valuations, and curbing inflows, with retrenchment of outflows during stress episodes (Caballero, Farhi, and Gourinchas, 2008a; Koepke, 2019; Davis, Valente, and van Wincoop, 2021).

Following Ludvigson et al. (2021), I set the threshold vector $\delta = [\delta_1, \delta_2, \delta_3, \delta_4, \delta_5, \delta_6]$ using the 75th percentile of the distribution obtained from random rotations of B in the unconstrained set. The estimated thresholds are $\delta = [2.90, 2.74, 2.52, 1.84, 2.16, 1.99]$, which discipline the admissible impulse responses.

The system of event constraints is written as

$$\bar{g}(\varepsilon_i(B), \delta) \geq 0, \quad (4)$$

where $\bar{g}(\cdot)$ stacks the inequality restrictions implied by Events $\bar{g}_1$ through $\bar{g}_7$.

External Variable Constraints

I use two external variables to further improve the identification of the model, leveraging their documented correlations with the shocks of interest. I center the choice on two stylized empirical facts. Gold and oil are standard in the literature for their relationships with macroeconomic and exchange rate uncertainty from a global perspective. Let $EX = [EX_g, EX_o]'$, where EX_g denotes gold return shocks and EX_o denotes oil supply news shocks. The external constraints are

$$\text{External 1 } (\overline{EC}_1) : \quad \rho(\varepsilon_i(B), EX_g) \geq 0,$$

$$\text{External 2 } (\overline{EC}_2) : \quad \rho(\varepsilon_i(B), EX_o) \leq 0,$$

where $\rho(\cdot, \cdot)$ denotes the sample correlation and $\varepsilon_i(B)$ is the structural shock implied by impact matrix B for component i .

These constraints exploit correlations between uncertainty and safe-asset or commodity shocks. The first constraint, $\overline{EC}_1$, captures the positive correlation between heightened uncertainty and demand for gold as a store of value during crises, consistent with Caballero, Farhi, and Gourinchas (2008b). The second constraint, $\overline{EC}_2$, uses the negative comovement between exchange rate fluctuations and adverse oil supply shocks, an empirical result in Känzig (2021). Oil and broader commodity shocks are well-known drivers of exchange rates and capital flows (Amano and van Norden, 1998; Chen and Chen, 2007; Basher, Haug, and Sadorsky, 2012; Chen, Liu, Wang, and Zhu, 2016). I therefore use these relationships to help identify the model's shocks through their expected signs.

I write the external constraint system compactly as

$$\bar{g}_C(\varepsilon_i(B), EX) \geq 0, \quad (5)$$

which stacks the inequalities implied by $\overline{EC}_1$ and $\overline{EC}_2$.

3 Data

I use multiple sources to estimate uncertainty shocks and their effects on capital flows. First, I take the daily Exchange Rate Uncertainty (ERU) measure of Fernandez-Mejia (2022) for 31 exchange rates spanning January 1999 to December 2019. I transform the data to a monthly frequency using the monthly median, which preserves within-month information better than end-of-month sampling and avoids averaging that can dilute spikes. Aggregating from daily to monthly also has identification advantages: Alessandri and Mumtaz (2019) shows that working directly at low frequency can introduce identification problems. The resulting dataset is a balanced panel with 249 observations per series.

To control for macroeconomic uncertainty, I use the monthly index of Jurado et al. (2015), which helps separate the exchange rate from domestic macro effects.⁵ Net Foreign Asset (NFA) flows come from the U.S. Treasury’s Treasury International Capital (TIC) system, which reports portfolio transactions between U.S. residents and foreigners in long-term securities.⁶ I seasonally adjust with the X-13ARIMA-SEATS procedure and difference series that exhibit unit roots to achieve approximate stationarity.⁷

The portfolio flow components include gross purchases of U.S. assets by foreigners (inflows) and gross purchases of foreign assets by U.S. residents (outflows). Asset classes are U.S. Treasury bonds, U.S. government agency bonds, U.S. corporate bonds, U.S. corporate stocks, foreign bonds, and foreign stocks. Following Gourinchas and Rey (2007, 2014); Atkeson et al. (2022), I define the NFA position as the market value of foreign assets held by U.S. residents minus the value of U.S. assets held by foreigners. I construct the NFA flow as foreign bonds and equities purchased by U.S. residents minus inflows, net of foreigners’ purchases of U.S. bonds and equities.

Figure 4 shows inflows, outflows, and the primary variable of interest, the NFA.⁸ Consistent with Gourinchas and Rey (2007) and Caballero et al. (2008a), inflows to the United States exceeded outflows from the early 2000s through the global financial crisis. The surge in inflows and subsequent sudden stop around late 2008 are embedded in the event restrictions

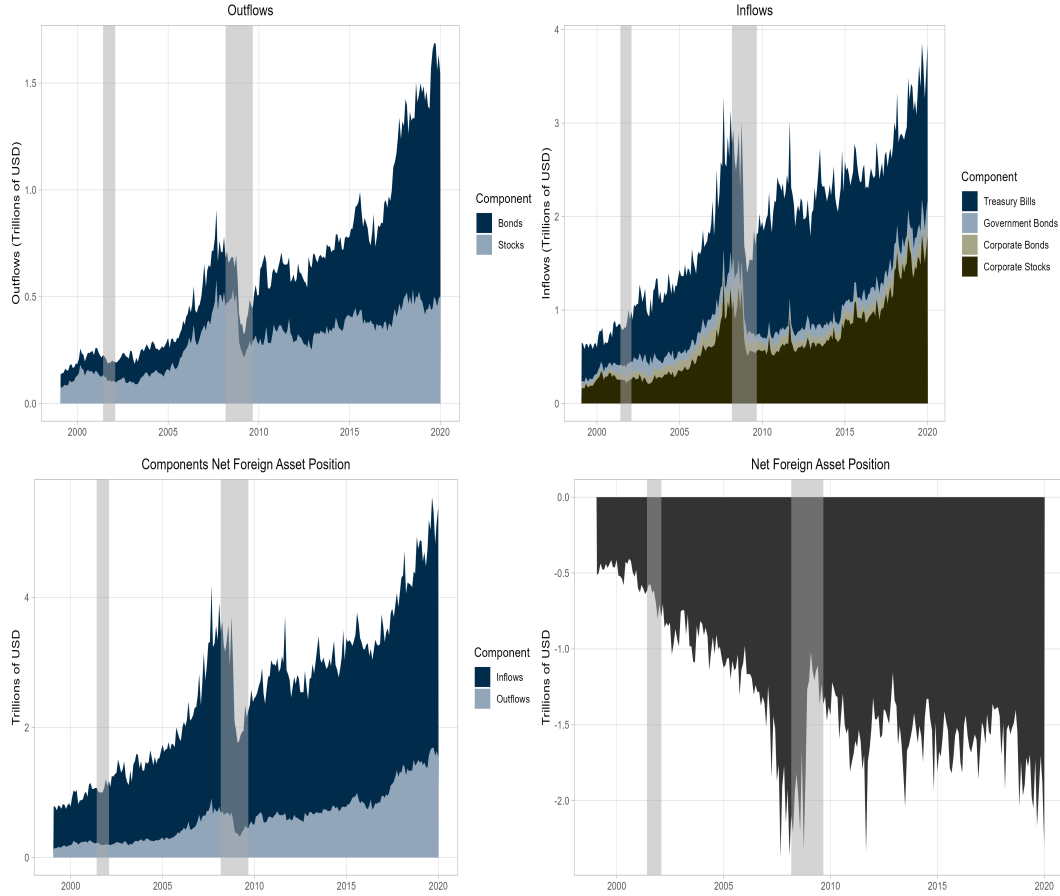
⁵Constructed from the FRED-MD database of McCracken and Ng (2016), comprising 134 U.S. macro-financial time series. I downloaded the 2020 vintage from Sydney Ludvigson’s webpage: <https://www.sydneyludvigson.com/macro-and-financial-uncertainty-indexes>.

⁶The TIC data cover portfolio transactions and exclude foreign direct investment (FDI). As emphasized by Koepke (2019) and Forbes and Warnock (2021), volatility in cross-border finance is predominantly reflected in portfolio flows, given FDI’s lower liquidity. Data: <https://home.treasury.gov/data/treasury-international-capital-tic-system-home-page/tic-forms-instructions/securities-a-us-transactions-with-foreign-residents-in-long-term-securities>.

⁷Table C1 in Appendix C reports descriptive statistics and unit root tests after seasonal adjustment.

⁸Appendix B decomposes the NFA into outflows of domestic investors and inflows from foreign demand for U.S. assets.

Figure 4. Stacked Inflows, Outflows, and Net Foreign Asset Position (USD trillions)

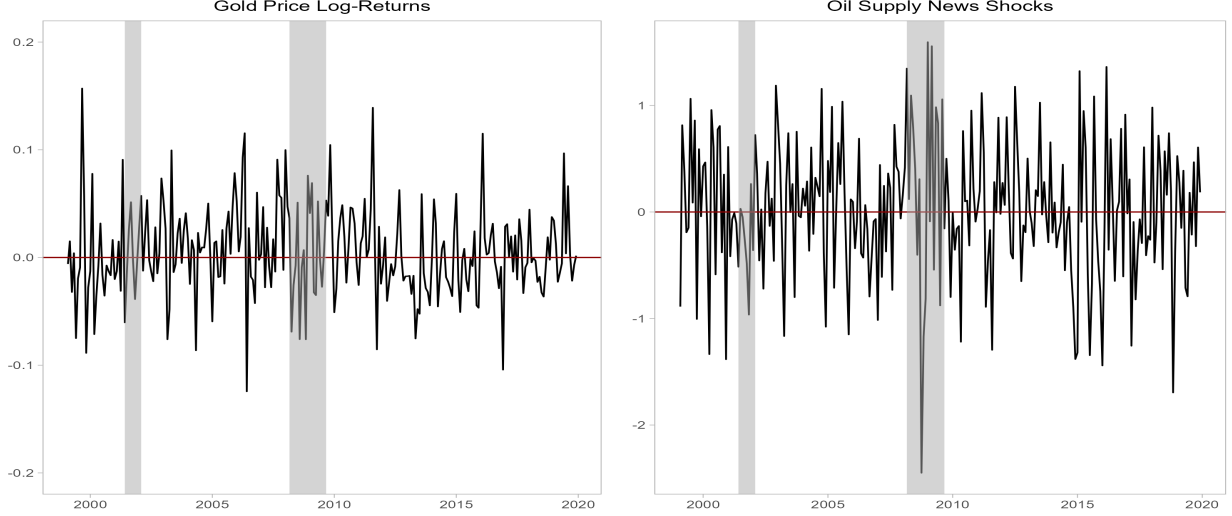


Note: Stacked time series for portfolio inflows, outflows, and the NFA position. The NFA flow is defined as U.S. residents' purchases of foreign assets minus foreigners' purchases of U.S. assets. *Sources:* TIC and author's calculations.

(e.g., $\bar{g}_5$). After the crisis, the NFA stabilized at a deficit of roughly \$1.5–\$2.0 trillion.

Gold prices are monthly USD per ounce from the London Bullion Market Association (LBMA), deflated by the U.S. Consumer Price Index (CPI) from FRED (base January 2018). The oil instrument is the oil supply news shock of Känzig (2021). Using shocks rather than raw oil returns separates structural oil supply disturbances from price movements due to other factors and yields a clearer mapping to exchange rates and risk premia. Figure 5 plots log returns of gold and the oil supply news shock, highlighting distinct volatilities and tail episodes. Notable periods include the 2007–2009 financial crisis and the negative oil supply shock that depressed prices from mid-2008 to early 2009.

Figure 5. External Instruments: Gold Log Returns and Oil Supply News Shocks



Note: Gold log returns are computed from LBMA monthly auction averages at 5 PM (USD/oz) and deflated by CPI (FRED) with base January 2018. Oil supply news shocks from Känzig (2021). *Sources:* LBMA, Känzig (2021), and author’s calculations.

4 Results

I present the results in two parts. First, I estimate the model focusing on the effects of macroeconomic and exchange rate uncertainty on the Net Foreign Asset (NFA) position. Second, I estimate the effects on inflows and outflows, with particular attention to the main contributors—bond and equity flows. Focusing only on net flows can obscure key mechanisms operating through the gross components.

4.1 Net Foreign Asset Position

Figure 6 reports impulse responses (IRFs) from variants of the VAR in Equation 1 that exclude narrative (event) restrictions. The endogenous vector is

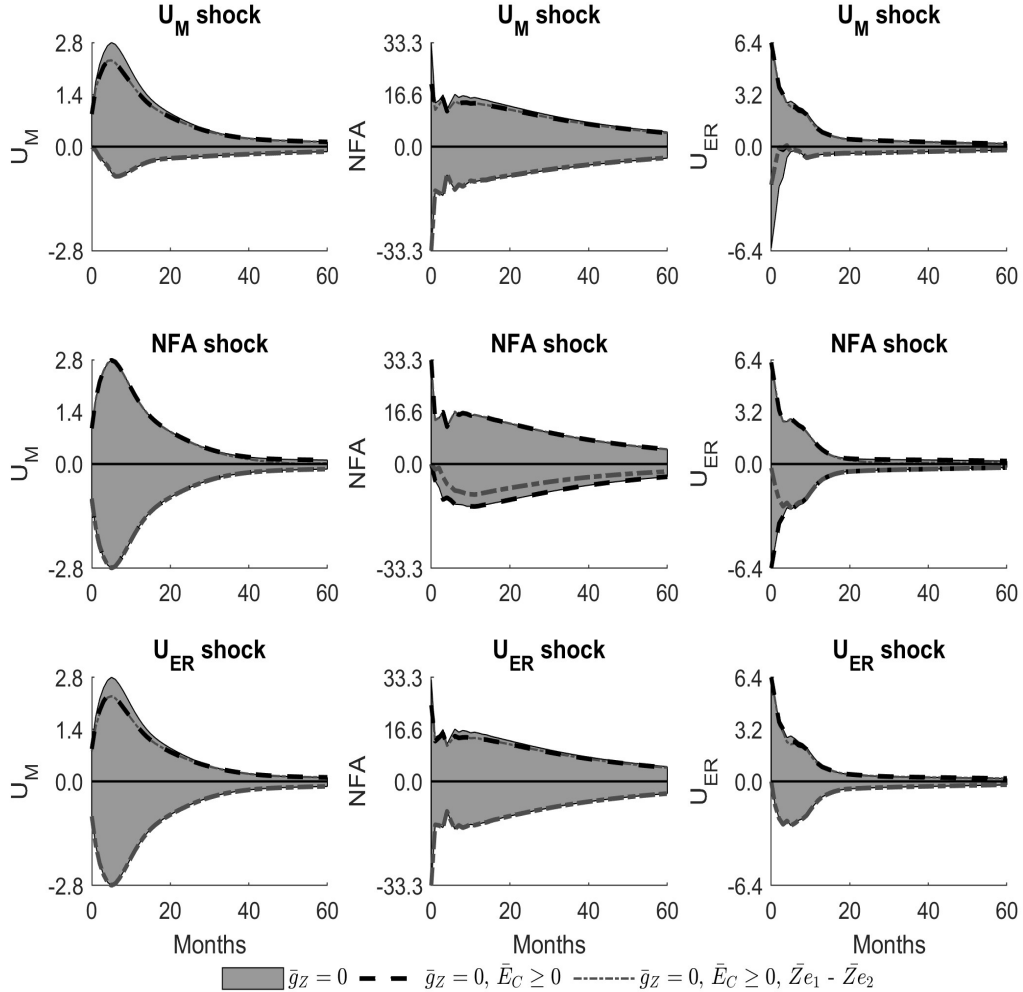
$$y_t = [U_{M,t}, NFA_t, U_{ER,t}]' \quad (6)$$

where $U_{M,t}$ is the one-month-ahead macroeconomic uncertainty index of Jurado et al. (2015), NFA_t is the United States NFA position, and $U_{ER,t}$ is the exchange rate uncertainty (ERU) index of Fernandez-Mejia (2022). I estimate the VAR with six lags: although standard information criteria favor two lags, residual diagnostics indicated persistent autocorrelation at shorter lag length.

The figure contrasts IRFs under three identification settings: (i) no additional restrictions

(covariance-matching only, $\bar{g}_\Sigma = 0$); (ii) sign restrictions on the external block, $\bar{g}_C \geq 0$; and (iii) external sign restrictions plus selected event sign restrictions, $\bar{g}_C \geq 0$ and $\bar{g}_{4,7}$. Columns indicate the shock source; rows report responses of each variable to that source.

Figure 6. Impulse Responses under Alternative Restrictions

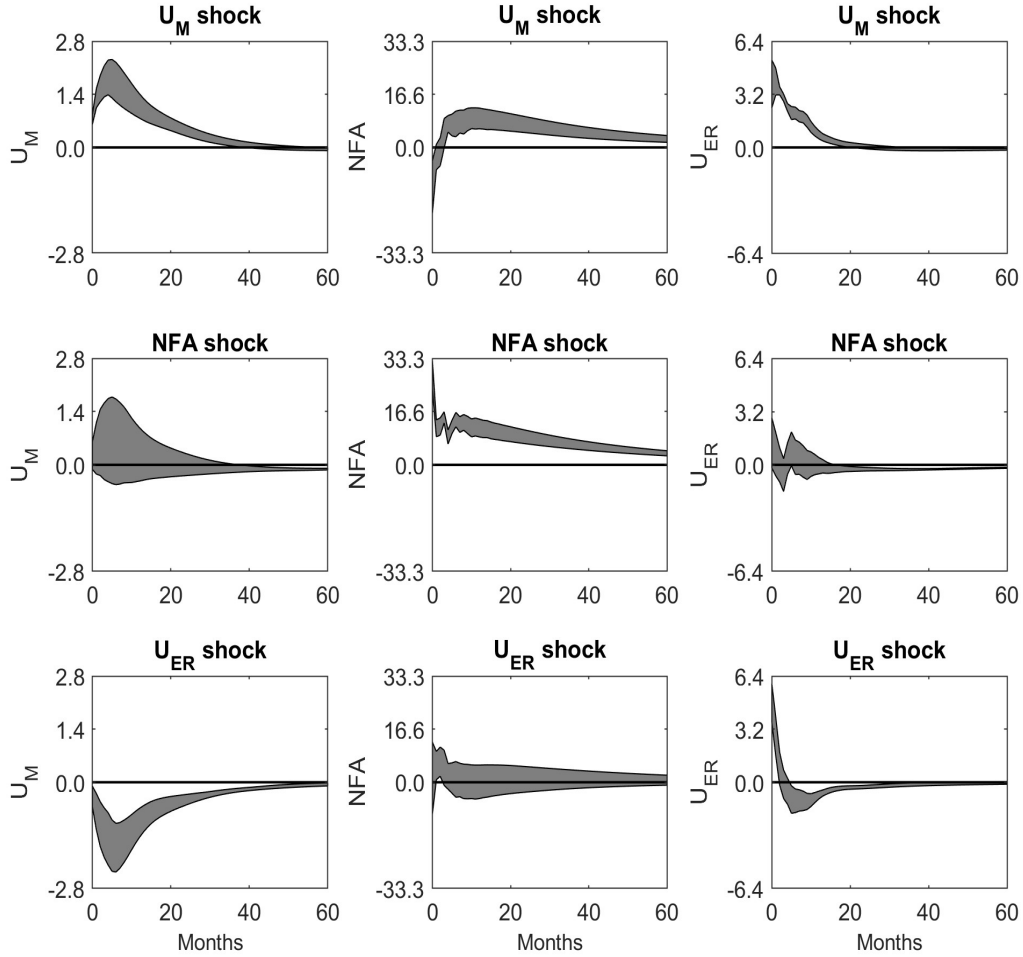


Note: IRFs from (i) reduced-form covariance matching only ($\bar{g}_\Sigma = 0$); (ii) with external sign constraints ($\bar{g}_C \geq 0$); and (iii) with external and selected event sign constraints ($\bar{g}_C \geq 0$, $\bar{g}_{4,7}$). Columns denote the shock; rows denote responses. NFA_t is measured in USD trillions. *Source:* Author's calculations.

Under covariance matching alone ($\bar{g}_\Sigma = 0$), IRFs are not informative—by construction, structural shocks are not identified. Adding $\bar{g}_C$ and $\bar{g}_{4,7}$ yields modest improvements by tightening the admissible rotation set, with a small positive effect of macroeconomic uncertainty shocks on exchange rate uncertainty. Overall, Figure 6 motivates the use of additional narrative constraints to sharpen the identification of the shocks of interest.

Figure 7 reports impulse responses from the specification that combines covariance match-

Figure 7. Impulse Response Functions Full Model



Note: IRFs from the fully disciplined design: covariance matching ($\bar{g}_\Sigma = 0$), external sign constraints ($\bar{g}_C \geq 0$), and narrative/event inequalities ($\bar{g}$). Columns denote the identified shock; rows denote responses. Shaded panels highlight the shock selection implied by the full set of constraints. NFA_t is measured in USD trillions. *Source:* Author's calculations.

ing ($\bar{g}_\Sigma$), external sign constraints ($\bar{g}_C$), and the full set of narrative/event inequalities ($\bar{g}$). The event set includes the September 2001 terrorist attacks ($\bar{g}_1$), the European sovereign debt crisis ($\bar{g}_3$), China's renminbi devaluation ($\bar{g}_5$), and the introduction of negative policy rates in Japan ($\bar{g}_6$). Shaded panels indicate the identified shock under the full model design. Relative to Figure 6, the comparison underscores the informational gains from targeted narrative constraints over sign-only schemes.

The IRFs indicate that a macroeconomic uncertainty shock initially deepens the NFA deficit, followed by a reversal and a sizable, persistent improvement at longer horizons. The

pattern is consistent with precautionary demand for domestic safe assets (see Figure B1 in Appendix B). Several channels may operate simultaneously. First, higher inflows into safe assets can crowd in demand for foreign assets by domestic investors (raising outflows). Second, a shift away from risky domestic assets can compress inflows, attenuating the NFA deficit. By contrast, exchange rate uncertainty shocks have limited effects on the NFA beyond a small, short-lived impact at the start of the horizon. ERU innovations are less pervasive than macroeconomic uncertainty shocks; exchange rate disturbances are absorbed relatively quickly, making their effects transitory. Taken together, the evidence suggests that policies that stabilize macroeconomic uncertainty can help mitigate NFA volatility and the persistence of external imbalances.

A useful feature of the framework is that it characterizes feedback across uncertainty types in addition to their effects on the NFA. Macroeconomic uncertainty increases exchange rate uncertainty on impact and over the near term, consistent with macro fragility that dampens both inflows and outflows. Given the United States' role as the dominant currency issuer—and the base for ERU construction—macro conditions naturally transmit to currency volatility. In turn, exchange rate uncertainty shocks are associated with a decline in macroeconomic uncertainty. The exchange rate channel operates through foreign-currency depreciation and terms-of-trade movements that reallocate portfolios toward safe assets, largely supplied by the United States (Camanho et al., 2022; Caballero, Farhi, and Gourinchas, 2016, 2021).

4.2 Gross Capital Inflows and Outflows

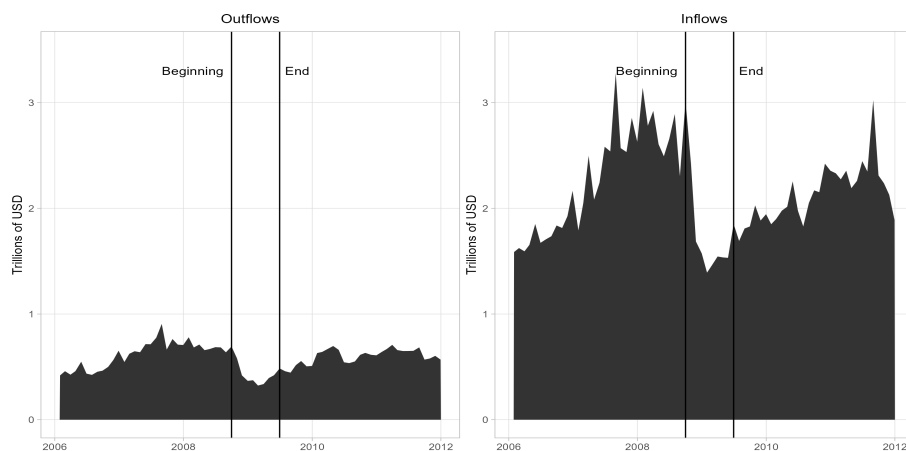
I now analyze the effects of uncertainty on gross capital inflows and outflows. Decomposing net flows clarifies the mechanisms driving movements in the NFA because offsetting changes in inflows and outflows can mask large underlying adjustments. To that end, I adapt the NFA restriction in Event $\bar{g}_7$ to the gross flow blocks and impose cumulative contraction during the crisis window:

$$\begin{aligned} \text{Event 8 } (\bar{g}_8) : & \quad \sum_{t \in \mathcal{T}} \varepsilon_{\text{OUT},t} \leq 0, \quad \mathcal{T} = \{\text{Sep 2008}, \dots, \text{Jun 2009}\}, \\ \text{Event 9 } (\bar{g}_9) : & \quad \sum_{t \in \mathcal{T}} \varepsilon_{\text{INF},t} \leq 0, \end{aligned}$$

where $\varepsilon_{\text{OUT},t}$ and $\varepsilon_{\text{INF},t}$ are the structural shocks to outflows and inflows, respectively. Figure 8 marks the start and end of the restriction window, matching the Lehman bankruptcy through the NBER recession trough in June 2009. Both inflows and outflows decline over

this period, with a steeper contraction in inflows. As seen in Figure 4, the post-crisis period features lower overall cross-border activity and a persistent contraction in the NFA. Subsequently, inflows recover more than outflows, worsening the NFA position—consistent with a reallocation toward domestic assets that erodes the United States’ privilege (Atkeson et al., 2022). Under higher macroeconomic and exchange rate uncertainty, elevated risk aversion reduces demand for U.S. risky assets and compresses capital inflows. At the same time, scarcity of safe assets in high-risk periods supports demand from international investors and central banks (Caballero et al., 2016), raising both inflows and outflows of bonds.

Figure 8. Restrictions on Capital Inflows and Outflows, 2006–2012

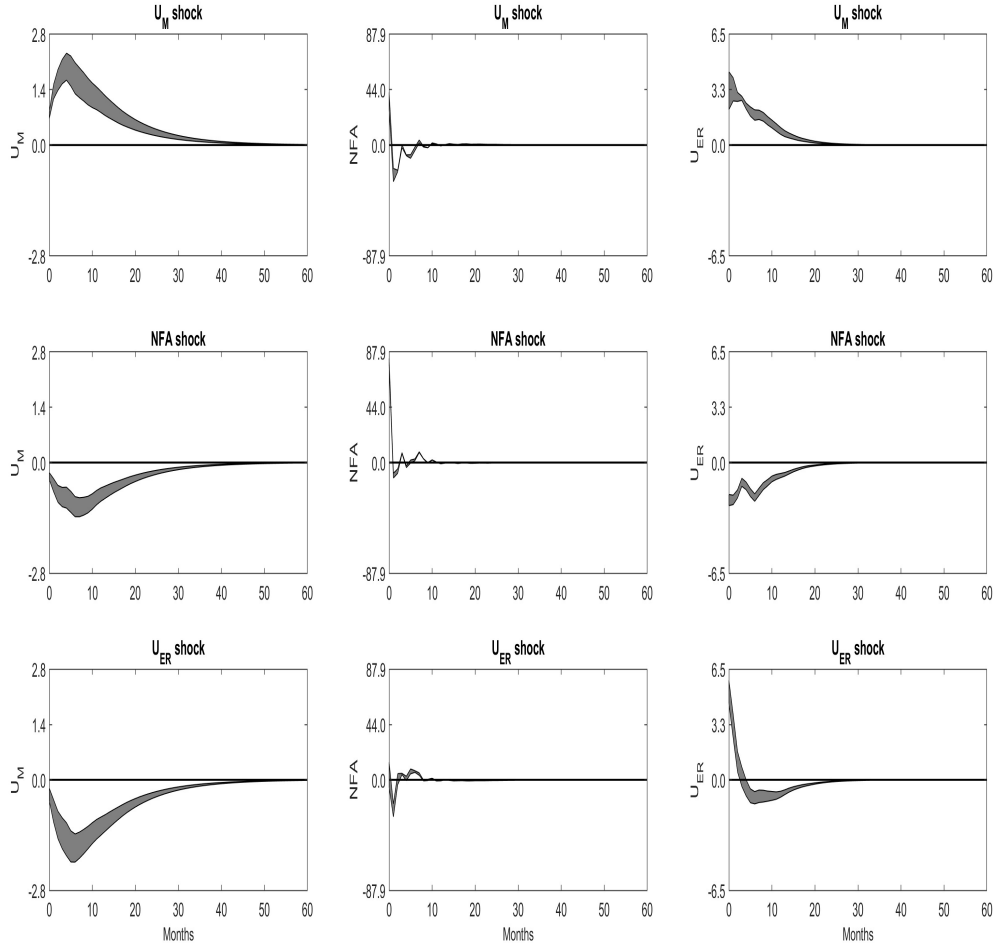


Note: Event restrictions $\bar{g}_8$ and $\bar{g}_9$ impose cumulative contractions of outflows and inflows between September 2008 and June 2009, as dated by the NBER Business Cycle Dating Committee. *Sources:* Author’s calculations and TIC data.

Figures 9 and 10 show impulse responses for outflows and inflows under the identification scheme used in the NFA VAR. Relative to the NFA specification, this setup is more restrictive, and the identified sets are narrower. Both uncertainty shocks reduce inflows and outflows on impact (second column in both figures). For outflows, macroeconomic uncertainty generates a negative response lasting about one year, whereas exchange rate uncertainty produces a more persistent contraction, on the order of two years. Thus, deterioration in U.S. macro conditions—coupled with heightened currency volatility—reduces domestic investors’ demand for foreign assets over the first year.

The outflow effects unwind sooner than exchange rate uncertainty normalizes, suggesting some tolerance to transient currency risk. Exchange rate shocks display a distinct pattern: as in the NFA model, ERU innovations lower macroeconomic uncertainty and initially depress outflows before turning positive. This is consistent with international shocks that raise the dollar, temporarily improving U.S. investors’ foreign purchasing power and prompting

Figure 9. Identified-Set Impulse Responses: Outflows

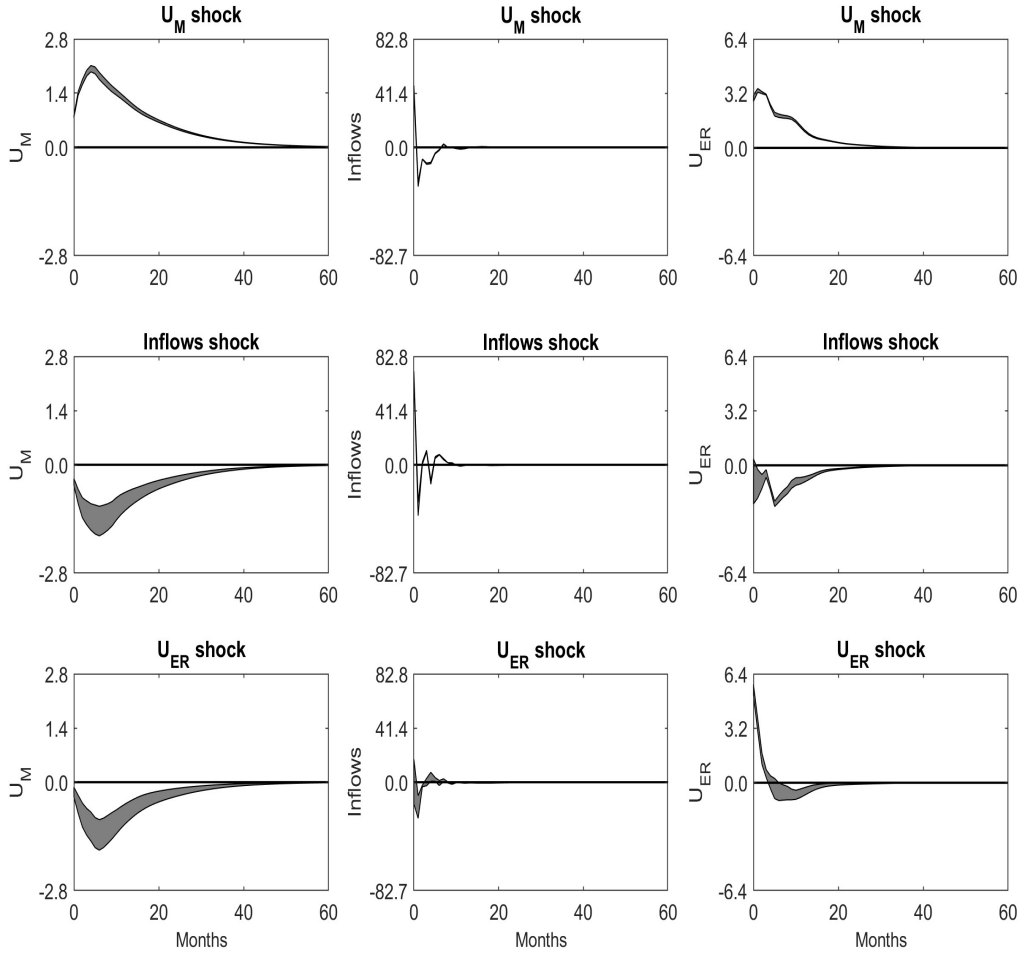


Note: IRFs from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author's calculations.

subsequent rebalancing toward foreign assets.

Inflows respond similarly in sign to outflows, but the magnitudes differ. Macroeconomic uncertainty raises exchange rate uncertainty and reduces inflows on impact; the contraction in inflows is more pronounced than in outflows, helping explain the positive NFA response at longer horizons in Figure 7. Exchange rate shocks also depress inflows, with smaller effects on outflows than inflows, which contributes to the improvement in the NFA position. In the sample considered, uncertainty shocks narrow the United States' external gap; macroeconomic uncertainty is quantitatively more important in closing the deficit.

Figure 10. Identified-Set Impulse Responses: Inflows



Note: IRFs from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author's calculations.

4.3 Components of Gross Portfolio Inflows and Outflows

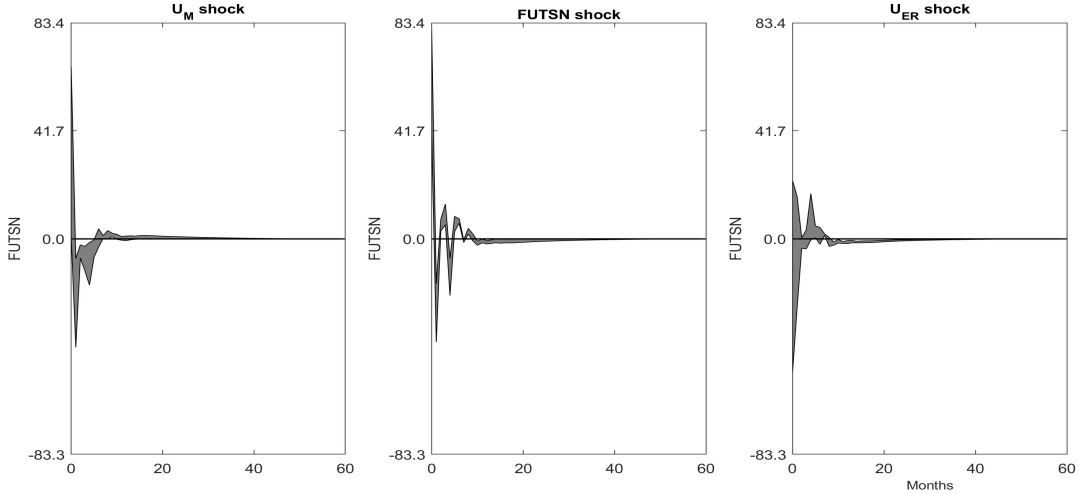
I decompose the largest components of gross flows—the demand for U.S. Treasury securities and U.S. corporate equities by foreigners, and U.S. demand for foreign bonds and foreign equities—to identify the sources of fluctuation in inflows and outflows. Heightened risk aversion should depress demand for risky assets (e.g., corporate equities), with stronger effects for macroeconomic uncertainty than for exchange rate uncertainty, since currency risk can be hedged and valuation effects may offset some USD movements. Thus, an increase in macroeconomic uncertainty is expected to reduce foreign demand for risky assets, whereas

exchange rate uncertainty should be more short-lived. I focus on the higher-value components and their determinants: U.S. Treasury bills and notes (bonds) and corporate stocks. I build on the previous identification and adapt the flow restrictions for each component:

$$\text{Event } j (\bar{g}_j) : \sum_{t \in \mathcal{T}} \varepsilon_{j,t} \leq 0, \quad \mathcal{T} = \{\text{Sep 2008}, \dots, \text{Jun 2009}\},$$

where $j \in \{\text{FUTSN}, \text{FUSC}, \text{UFB}, \text{UFSC}\}$ indexes (i) Foreign demand for U.S. Treasury bills and notes (FUTSN), (ii) Foreign demand for U.S. corporate stocks (FUSC), (iii) U.S. demand for foreign bonds (UFB), and (iv) U.S. demand for foreign corporate stocks (UFSC). This component-level analysis speaks directly to the evolution of the United States’ “exorbitant privilege” and “duty” (Gourinchas and Rey, 2007, 2014, 2022). Figures 11 and 12 report identified-set impulse responses (IRFs) for bond-related inflows and outflows, represented by FUTSN and UFB.

Figure 11. IRFs: Foreign Demand for U.S. Treasury Bills and Notes (FUTSN)

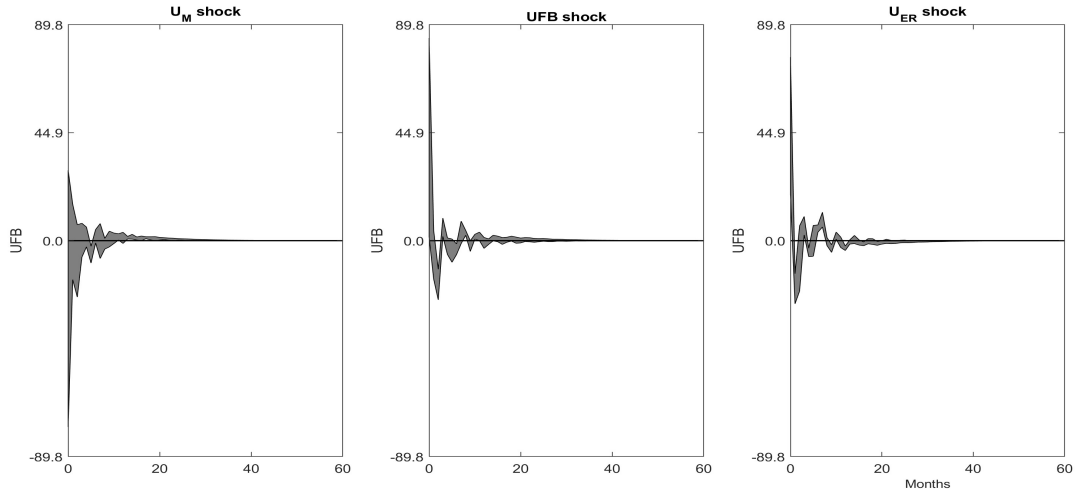


Note: IRFs from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author’s calculations.

The effects differ across bond inflows and outflows. An increase in macroeconomic uncertainty reduces foreign demand for U.S. bonds (FUTSN). By contrast, I find no strong evidence that macroeconomic uncertainty alters U.S. demand for foreign bonds (UFB). This asymmetry is consistent with a heightened perception of U.S. macro risk that, despite the United States’ role as a safe-asset issuer, temporarily reduces foreign purchases. Valuation channels can reinforce this mechanism: tighter financial conditions raise yields and depress bond prices, lowering the value of foreign portfolios even if quantities adjust modestly.

Bond outflows (UFB) are less connected to uncertainty shocks, plausibly reflecting domestic portfolio rebalancing that diversifies sovereign risk with other foreign safe assets. Given that bonds account for a large share of cross-border portfolios, NFA movements are more sensitive to changes in inflows than outflows. Exchange rate uncertainty shows the opposite pattern: increases in ERU do not materially affect foreigners’ demand for U.S. Treasuries (FUTSN)—consistent with the insurance role of U.S. safe assets during stress—whereas ERU depresses U.S. demand for foreign bonds (UFB) on impact, followed by a recovery. Higher exchange rate volatility lowers foreign-currency returns on coupons and principal and is associated with larger deviations from covered interest parity (Kalemli-Özcan and Varela, 2021; Fernandez-Mejia, 2022), dampening outbound bond purchases initially.

Figure 12. IRFs: U.S. Demand for Foreign Bonds (UFB)

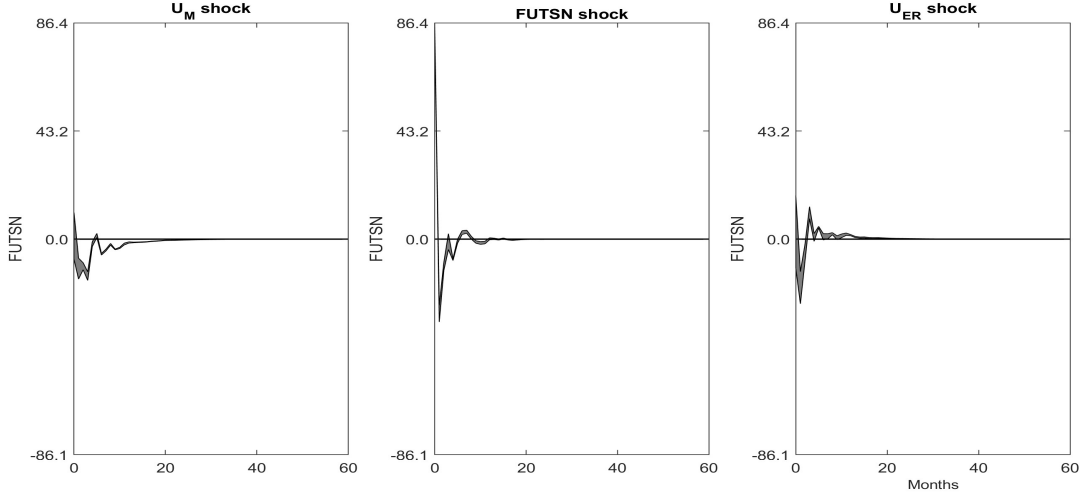


Note: IRFs from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author’s calculations.

I compare corporate equities to bonds to isolate which components drive fluctuations in gross flows. The equity response is broadly symmetric across inflows and outflows, with larger effects on outflows than inflows. Macroeconomic uncertainty shocks initially reduce U.S. residents’ demand for foreign equities, consistent with “wait-and-see” behavior and deferral of investment decisions (Baker et al., 2016). The domestic (macroeconomic) channel is stronger and more persistent than the exchange rate channel. Exchange rate uncertainty also lowers gross equity flows on impact, followed by a rebound over several periods. Overall, for equities, both uncertainty types produce qualitatively similar responses, with differences in persistence and magnitude.

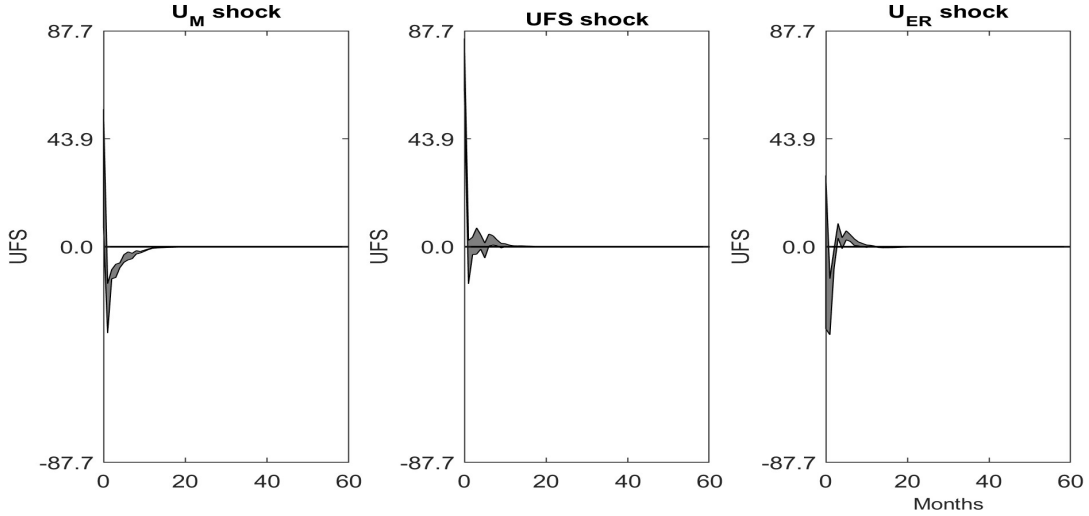
Taken together, the components of gross portfolio flows exhibit heterogeneous and asym-

Figure 13. IRFs: Foreign Demand for U.S. Corporate Stocks (FUSC)



Note: Identified-set impulse responses from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author's calculations.

Figure 14. IRFs: U.S. Demand for Foreign Corporate Stocks (UFSC)



Note: Identified-set impulse responses from (i) covariance matching only ($\bar{g}_\Sigma = 0$), (ii) with external sign constraints ($\bar{g}_C \geq 0$), and (iii) with external plus narrative/event constraints ($\bar{g}_C \geq 0, \bar{g}$). Columns denote the identified shock; rows denote responses. *Source:* Author's calculations.

metric responses to macroeconomic and exchange rate uncertainty. Inflows are more sensitive than outflows, helping reconcile the positive NFA response in Figure 7. The transfer effect is similar to Gourinchas and Rey (2022) in the post-crisis period. Exchange rate uncertainty effects on equities are often larger on impact than macroeconomic uncertainty but less per-

sistent, consistent with the shock-absorption role of exchange rates. Comparing uncertainty series in Figure 2, macroeconomic shocks are less frequent but longer-lasting. Consistent with reduced investment during uncertainty episodes, the United States’ privilege is attenuated as foreign demand for U.S. bonds and equities falls. The United States also reduces its NFA position during uncertainty spikes; this reduction is smaller under exchange rate shocks alone. When both uncertainty types are elevated, the combined effect is larger, aligning with the “exorbitant duty” narrative: the loss of external position raises the market value of foreign liabilities and reallocates liquidity toward the rest of the world.

5 Conclusions

I examine how exchange rate and macroeconomic uncertainty shape the United States’ Net Foreign Asset (NFA) position. The analysis employs a VAR estimated under complementary identification schemes—sign restrictions, narrative (event) inequalities, and external-variable constraints—so that uncertainty innovations are disciplined by high-uncertainty episodes. This design separates macroeconomic from exchange rate uncertainty shocks and identifies their distinct effects.

The estimates reveal asymmetric interactions between the two forms of uncertainty: macroeconomic uncertainty raises exchange rate volatility, whereas exchange rate uncertainty does not meaningfully amplify macroeconomic uncertainty. Shocks to the NFA, in turn, have negligible feedback on either uncertainty measure.

NFA responses also differ by uncertainty type. Macroeconomic uncertainty initially deepens the NFA deficit but subsequently reverses into a sizable and persistent improvement, consistent with heightened demand for domestic safe assets and portfolio rebalancing. Exchange rate uncertainty shocks have small and short-lived effects on the NFA, reflecting the shock-absorption role of exchange rates. Disaggregating net flows clarifies the mechanism: inflows are more sensitive than outflows, accounting for much of the NFA response.

At the component level, bonds and equities display distinct patterns. Macroeconomic shocks reduce foreign demand for United States safe assets (Treasuries) and depress equity flows; exchange rate uncertainty has little effect on foreign demand for Treasuries but dampens United States purchases of foreign bonds on impact. For equities, both uncertainty types depress foreign demand, with exchange rate effects larger on impact but less persistent. These results align with the “privilege–duty” framework: under uncertainty shocks, the United States’ privilege weakens while its duty persists as global investors continue to demand safe assets. The United States thus retains the benefits of sustained demand for its liabilities but faces rising costs associated with debt sustainability.

The findings provide inputs for calibrating models that embed uncertainty shocks into capital flow dynamics and the evolution of the NFA. Future work could further separate the roles of exchange rate volatility and domestic macro shocks in driving gross capital movements. From a policy perspective, measures that stabilize macroeconomic uncertainty can mitigate NFA volatility and the persistence of external imbalances; the evidence for exchange rate uncertainty is weaker and less persistent.

Two limitations merit emphasis. First, the data aggregate portfolio transactions, complicating the separation of valuation changes from quantity flows (Gourinchas and Rey, 2007, 2014; Atkeson et al., 2022). Incorporating investor heterogeneity, as in Camanho et al. (2022), could clarify portfolio rebalancing channels under exchange rate shocks. Second, identification is set- rather than point-based, yielding wide confidence intervals and limiting parameter precision. Policy conclusions should therefore be interpreted with appropriate caution.

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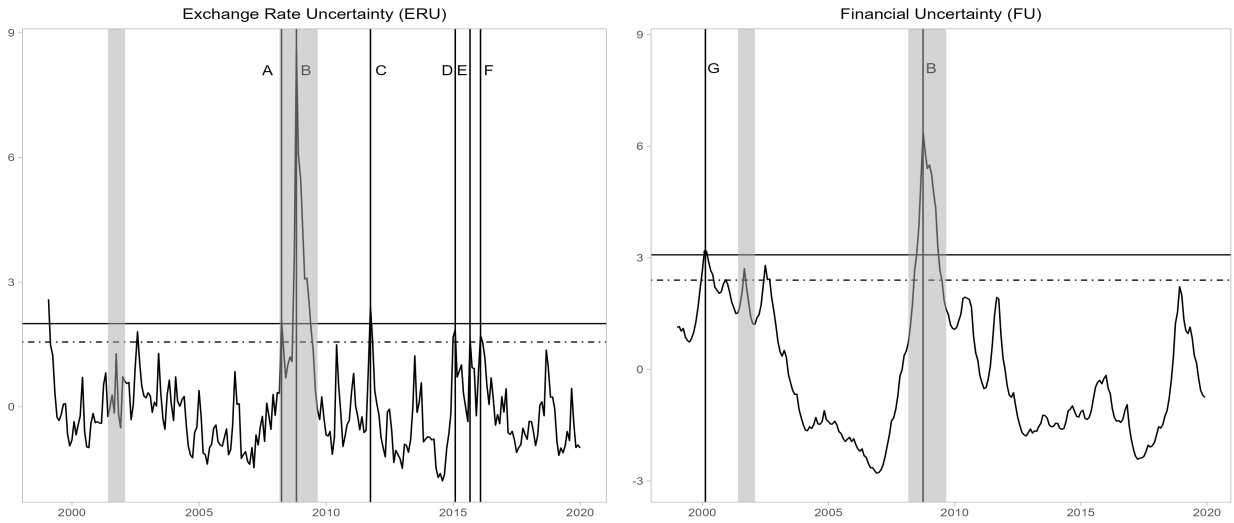
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Appendix

A Uncertainty Comparisons

In Figure A1, I show the calculated for the ERU and the Financial Uncertainty of Ludvigson et al. (2021). Without looking at the events, it is clear that the behavior of both variables is different. The episodes of financial Uncertainty are given by: F is the dot-com crisis in 2000; G is the 9/11 terrorist attacks; H is the second market correction posterior to the crisis; I is the Lehman Default. None of the first three events is significant in the ERU.

Appendix Figure A1. Exchange Rate and Financial Uncertainty

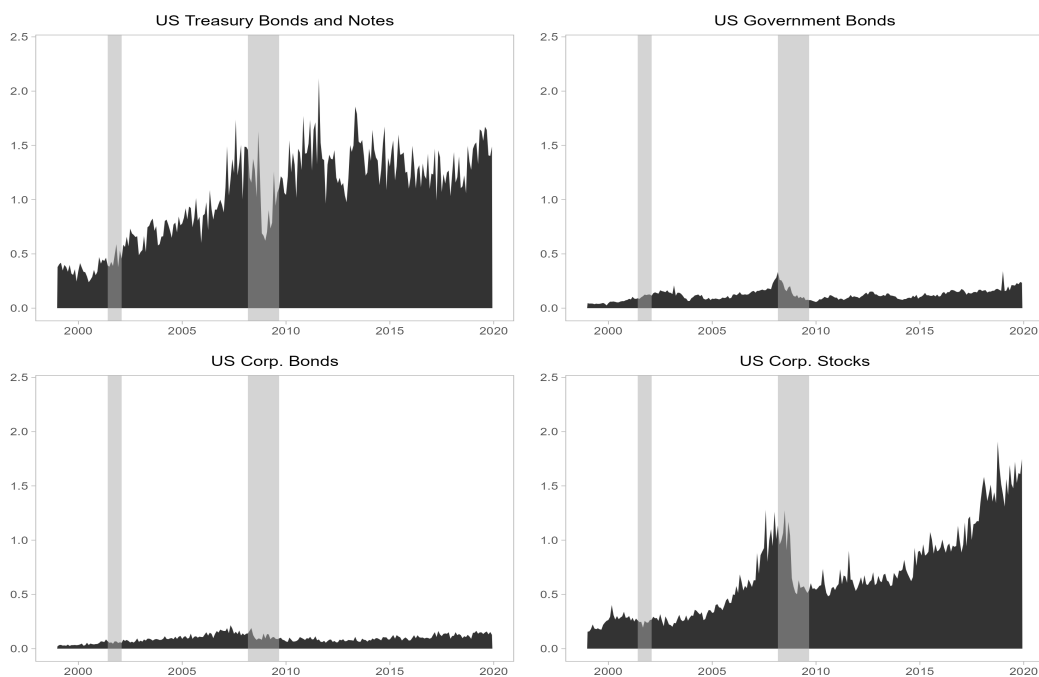


Note: The graph presents the ERU and Financial Uncertainty with the mean of the series standardized to zero. The horizontal line corresponds to the 1.65 standard deviations above the mean, defining each series's important episodes. The vertical line is the peak period of each period identified and used as the event constraint. *Sources:* Ludvigson et al. (2021), Fernandez-Mejia (2021), and Author own calculations.

B Net Foreign Asset Position Composition

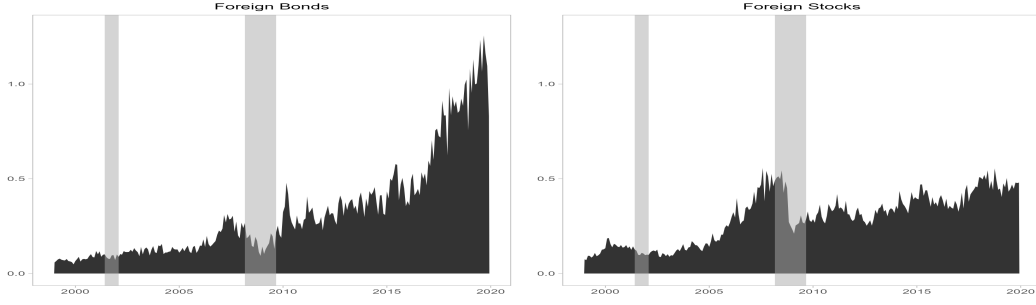
In Figures B1 and B2, I present the components of the Net Foreign Asset Position of the United States that define the gross portfolio capital inflows and outflows. The inflows are the purchases of US Treasury Bonds and Notes, US Government Bonds, US corporate bonds, and US corporate stocks. At the same time, the Outflows consist of the demand for Foreign Bonds and Foreign stocks. By looking at the magnitudes, we can conclude that foreigners' higher demand for US assets is related to the Treasury Bonds and Stocks, consistent with the literature on NFA and the Exorbitant Privilege. In comparison, the outflows go consistently to foreign bonds, which we can presume are for diversification motives and because of their high yields.

Appendix Figure B1. Gross Outflows of Portfolio Capital Flows



Sources: TIC

Appendix Figure B2. Gross Outflows of Portfolio Capital Flows



Sources: TIC

C Descriptive Statistics

Table C1 presents the descriptive statistics of the raw data gathered to estimate the model of Equation 1. The whole data set includes the Uncertainty of interest, the price of gold, and the Oil Supply news shocks. The other variables correspond to the components of the portfolio inflows and outflows of the series. One part of the components is the gross purchases of the Foreigners of US Treasury Bills and Notes (FUSTN), Government Bonds (FUSB), Corporate Bonds (FUSCB), Corporate Stocks (FUSS), Foreign Bonds (FFB), and Foreign Stocks (FFS) which are the Inflows. The other part is the gross sales of foreigners to the US residents of US Treasury Bills and Notes (UUSTN), Government Bonds (UUSB), Corporate Bonds (UUSCB), Corporate Stocks (UUSS), Foreign Bonds (UFB), and Foreign Stocks (UFS) which are the Outflows. Finally, the Net Foreign Assets are defined as:

$$NFA = (UFS + UFB) - (FUSTN + FUSB + FUSCB + FUSS + FFB)$$

We can see that based on the Augmented Dickey-Fuller, the last column in the Table C1, the variables of interest U_M , U_{ER} , and Oil Supply do not need to be transformed as we find no evidence of the presence of unit root in the series. This is not the case with the Gold price, to which I calculate the logarithmic Returns.

Table C2 presents the Lag selection of the VAR model such that the optimal number of lags of the VAR model minimizes the majority of the criteria. We can see that according to each, the best model for the estimation is that it uses two lags. Nevertheless, it still presents some level of serial correlation in the errors. Then, I use the more parsimonious model with no evidence of serial correlation, the one with six lags.

Appendix Table C1. Descriptive Statistics

Variable	Min	Q1	Median	Mean	Q3	Max	St.Dev.	Range	ADF
Unc. Macro	0.55	0.61	0.64	0.66	0.69	1.08	0.09	0.08	-3.36
Unc. ER	0.35	0.46	0.51	0.53	0.57	1.39	0.12	0.12	-5.17
gold.inf	373.11	543.50	1138.01	1036.49	1362.01	2080.20	472.07	818.50	-1.14
oil.shocks	-2.45	-0.37	0.01	0.02	0.44	1.59	0.65	0.82	-10.92
FUSTN	0.25	0.72	1.17	1.04	1.35	1.91	0.40	0.63	-1.90
FUSB	0.03	0.09	0.12	0.12	0.15	0.34	0.05	0.06	-2.53
FUSCB	0.02	0.07	0.09	0.10	0.12	0.20	0.03	0.04	-2.45
FUSS	0.16	0.32	0.60	0.68	0.92	1.81	0.39	0.59	-0.35
FFB	0.06	0.12	0.24	0.33	0.40	1.22	0.28	0.28	1.02
FFS	0.08	0.15	0.31	0.29	0.39	0.56	0.13	0.24	-1.32
UUSTN	0.25	0.70	1.15	1.03	1.34	1.90	0.40	0.64	-1.89
UUSB	0.02	0.08	0.10	0.11	0.13	0.32	0.05	0.05	-2.58
UUSCB	0.01	0.07	0.08	0.08	0.10	0.18	0.03	0.04	-1.71
UUSS	0.16	0.32	0.59	0.68	0.92	1.83	0.40	0.60	-0.43
UFB	0.06	0.12	0.26	0.33	0.39	1.21	0.27	0.27	0.68
UFS	0.07	0.15	0.32	0.30	0.39	0.58	0.13	0.24	-1.46
FI	0.58	1.26	2.05	1.94	2.50	3.86	0.79	1.23	-0.99
USO	0.14	0.26	0.61	0.62	0.78	1.69	0.38	0.52	0.41
NDA	-2.37	-1.63	-1.41	-1.32	-0.98	-0.41	0.47	0.65	-2.15

Notes: The table presents different statistics for each of the 19 variables used in constructing the model. Min is the minimum, Q1 is the first quartile, Q3 is the third quartile, Max is the Maximum, St. Dev. is the Standard deviation of the series, Range is the Interquartile difference between Q3 and Q1, and ADF is the test statistic of the Augmented Dickey-Fuller test with drift. For the ADF test, the null hypothesis is that the series has a unit root. There are 249 data points, so the ADF test's critical value is -2.87.

Appendix Table C2. Lag Selection of the VAR Model

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
AIC	-6.69	-7.62	-7.62	-7.62	-7.58	-7.55	-7.50	-7.49	-7.43	-7.40	-7.37	-7.33	-7.27	-7.24
HQ	-6.62	-7.50	-7.45	-7.39	-7.30	-7.21	-7.11	-7.05	-6.94	-6.85	-6.77	-6.68	-6.56	-6.48
SC	-6.51	-7.32	-7.18	-7.05	-6.88	-6.72	-6.54	-6.39	-6.20	-6.04	-5.87	-5.71	-5.51	-5.36
FPE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

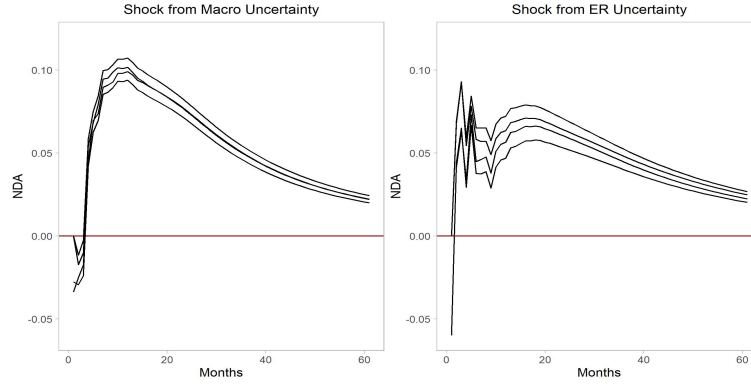
Notes: The table presents the different information criteria for selecting the number of lags for the VAR model. The optimal lag model should be such that it minimizes each.

D Multiple Specifications of VAR

I estimate different combinations of the VAR model using the recursive identification by Cholesky decomposition to contrast the robustness of the model. The identification requires imposing contemporaneous relationships between variables and response delays between others. As there is no framework to define a strict order in which to accommodate the endogenous variables, I will estimate all the possible permutations of the vector of variables $X_t = [U_{M,t}, NFA_t, U_{ER,t}]'$. This implies estimating 2^3 VAR models and obtaining their respective IRFs. In Figure D1, I present the result of estimating two a shock of each Uncertainty shock on the Net Foreign Asset Position.

The result of the VAR shows consistency in the relationship indifferent to the ordering of the variables. The change in the different models reflects an order of magnitude rather than an effect change. The main difference with the results of Figure 7 is the initial effect of the

Appendix Figure D1. Recursive VAR



Note: The Figure presents the orthogonal IRFs from the Macroeconomic and Exchange Rate Uncertainty shocks in the NFA under different models. Each line represents the mean IRF of one of the six models constructed from the permutation of the three variables that determine the recursive ordering identification. The calculated IRF error bands (not shown) are derived by bootstrapping with a thousand runs. *Sources:* Author's Calculations

shock in the NFA. In the VAR, the initial effect is a modest, non-significant decrease with a later significant increase in the position's value. Due to the imposed narrative restrictions, I identify a substantial shock from macroeconomic and exchange rate uncertainty, the highest and most persistent for the first. As we see episodes like the financial crisis of 2008, we can compare the behavior of the uncertainties and NFA in Figures 2 and 4, where we can see that the increase in Uncertainty is related to episodes of lower positions. The difference highlights the advantage of the identification scheme used over the short and long-run identification restrictions in the VAR model.